forecast report sample airline passengers

1. Executive Dashboard

Metric	Value	Status
↗ Forecast Pattern	Seasonal / Variable	info
▥ Forecast Endpoint Change	-2.0%	negative
◎ Forecast Confidence	65/100 — Medium	warning
◉ Data Quality	Fair	warning
◆ Model Selected	Holt-Winters	info
⚠ Primary Risk	Candidate structural breaks require validation	warning
✓ Recommended Action	Review statistical audit findings and monitor future actuals	warning

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2. Executive Summary

Strategic Outlook: The forecast follows a seasonal / variable path over the 12-period horizon and ends at 432.28, compared with 441.11 in the first forecast period.

Forecast Endpoint Change: -2.0% from the first forecast period to the last

Confidence Level: 65/100 — Medium

Primary Risk: Forecast accuracy may decline over longer horizons

Recommended Action: Review the statistical audit findings and monitor forecast performance against future actuals.

The seasonal, variable trajectory over the next twelve periods begins just above the low-440 range and settles just under the mid-430 mark, indicating a modest shift across the horizon. Confidence sits at a medium level, reflected by a 65-point rating, because the model's performance tends to soften as the forecast extends further out. The chief concern is that predictive accuracy can erode with longer lead times, potentially widening the forecast envelope. Accordingly, the team should audit the statistical findings and keep a close watch on how actuals align with the projections.

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3. Data Quality Summary

Overall Rating: Fair

Collection quality is good under the completeness and regularity policy: 0 missing values, 0 duplicate timestamps, and 0 gaps; 0 validation issues; intervals are regular. Anomaly risk requires review: 11 detected values (7.6%) exceed the 5% review threshold, limiting the overall rating to Fair.

Metric	Value
Missing Values	0
Duplicate Timestamps	0
Missing Timestamps (Gaps)	0
Outlier Count	11
Outlier Ratio	7.6%
Regular Intervals	Yes
Frequency	MS
Completeness	100.0%

The data receives a Fair rating because completeness is perfect—100 % of the expected monthly timestamps are present with no missing values, duplicates, or gaps—and the series is fully regular. The rating is limited by anomaly risk: eleven observations, or 7.6 % of the points, exceed the 5 % review threshold, which is the most significant issue identified. This elevated anomaly proportion can introduce uncertainty into forecasts, so ongoing monitoring and possible data-cleansing may be needed to preserve forecast reliability.

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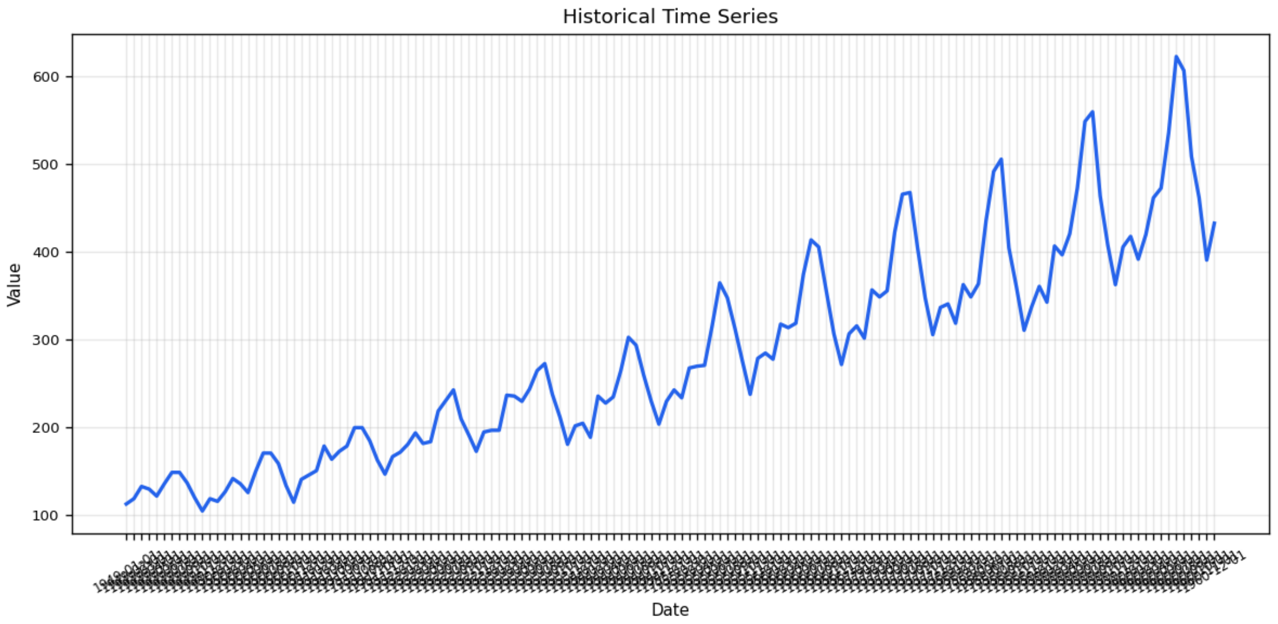
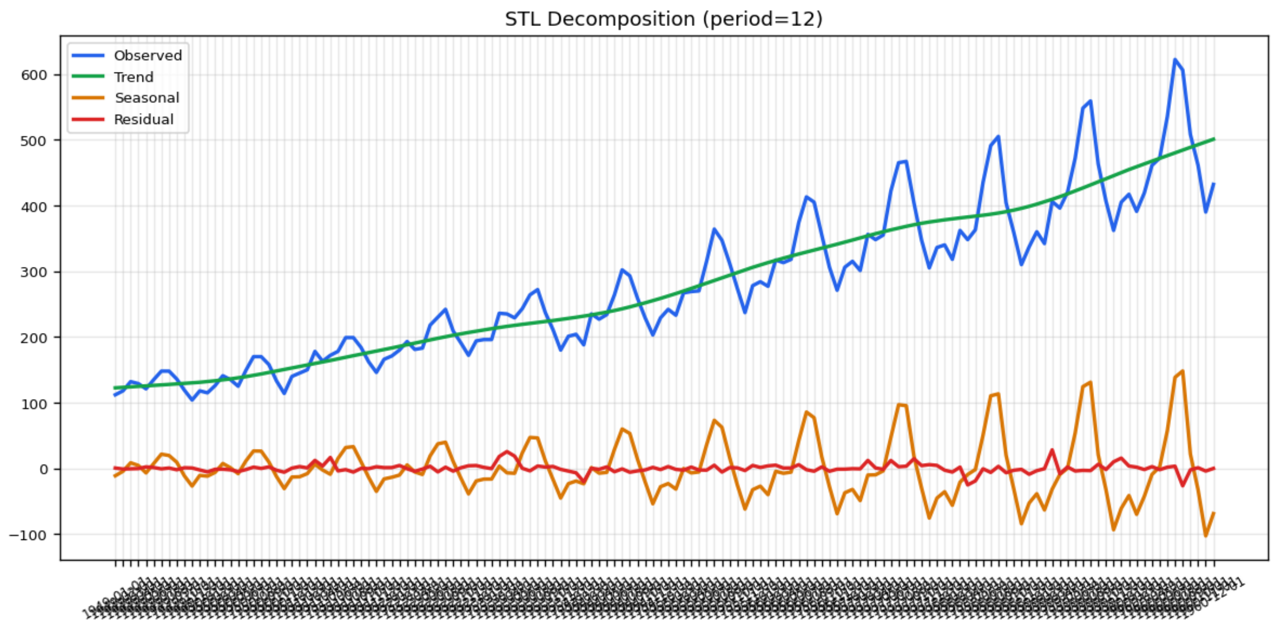
4. Historical Performance & Trend Analysis

Trend Direction: Upward
Trend Detected: Yes
Seasonal Pattern: Every 12 periods
Statistical Stability: Changing over time

The data show a clear upward movement, with the metric rising by roughly 2.66 units each month. This steady increase signals growing demand that may warrant a review of operational capacity to sustain momentum. A recurring twelve-month cycle is evident, indicating that performance peaks and troughs align with the calendar year. Monitoring these seasonal swings will help align resources with periods of higher activity.

Figure: Historical Data
The historical time series showing the underlying trend and seasonal patterns.





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5. Future Growth & Forecast Outlook

Forecast Horizon: 12 periods (1961-01-01T00:00:00 → 1961-12-01T00:00:00)
Endpoint Change: -2.0%
Endpoint Direction: Downward
Forecast Pattern: Seasonal / variable
Seasonal Peak: 615.9965 on 1961-07-01T00:00:00 (+39.6% versus the first forecast period)
Start Value: 441.1108

End Value: 432.2808
Metric Provenance: Model selection used rolling-origin metrics; the untouched final-test metrics below were not used for ranking.
Untouched Final Test: RMSE 29.7418, MAE 26.7118
Recent Holdout Assessment: Recent untouched final-test RMSE was 1.00× the pooled rolling-origin RMSE, broadly consistent with the rolling validation evidence.

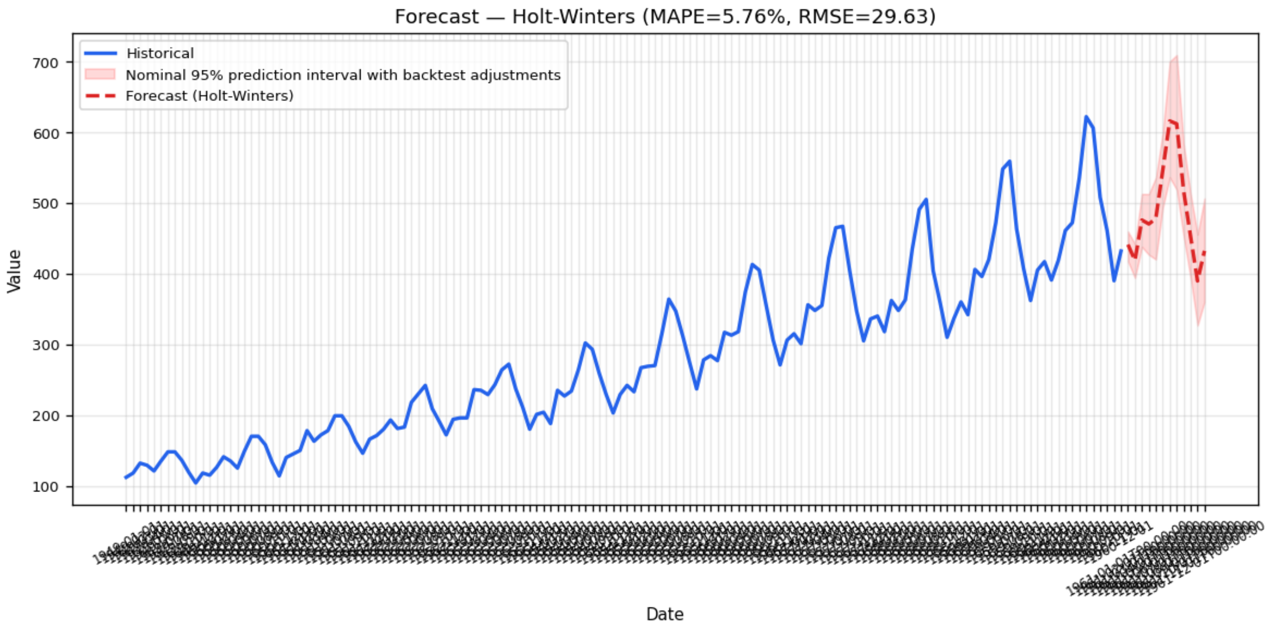
The Holt-Winters model projects a seasonal / variable pattern across the 12-month horizon. The series moves from 441.1 at the start to 432.3 at the end, a –2.0 % shift over the period. A temporary peak of 616.0 is forecast for July, a 39.6 % rise relative to the start, after which values revert toward the endpoint level. All monthly forecasts include a 95 % prediction interval (e.g., January 417.4–459.6, July 536.3–700.0), highlighting the uncertainty that should be monitored as actuals arrive.

95% Prediction Intervals with Backtest Adjustments

Backtest interval adjustment applied at 12 of 12 horizons. Each adjusted horizon uses at least five non-overlapping observed backtest errors. Other horizons retain their original model ranges. This empirical correction does not guarantee 95% coverage.

Date	Forecast	Lower Bound	Upper Bound
1961-01-01T00:00:00	441.1108	417.3886	459.5965
1961-02-01T00:00:00	418.6482	394.0341	442.2434
1961-03-01T00:00:00	475.8417	437.82	512.7613
1961-04-01T00:00:00	470.311	426.74	512.5326
1961-05-01T00:00:00	477.88	420.2341	535.7228
1961-06-01T00:00:00	546.0383	493.2754	599.4197
1961-07-01T00:00:00	615.9965	536.2505	700.0383
1961-08-01T00:00:00	611.9267	518.77	709.2263
1961-09-01T00:00:00	517.1572	452.8489	584.0706
1961-10-01T00:00:00	452.6195	393.9857	513.0596
1961-11-01T00:00:00	389.676	326.7993	455.6341
1961-12-01T00:00:00	432.2808	360.2932	506.4593

Figure: Forecast with Backtest-Adjusted Prediction Intervals
The projected values with a nominal 95% planning range.



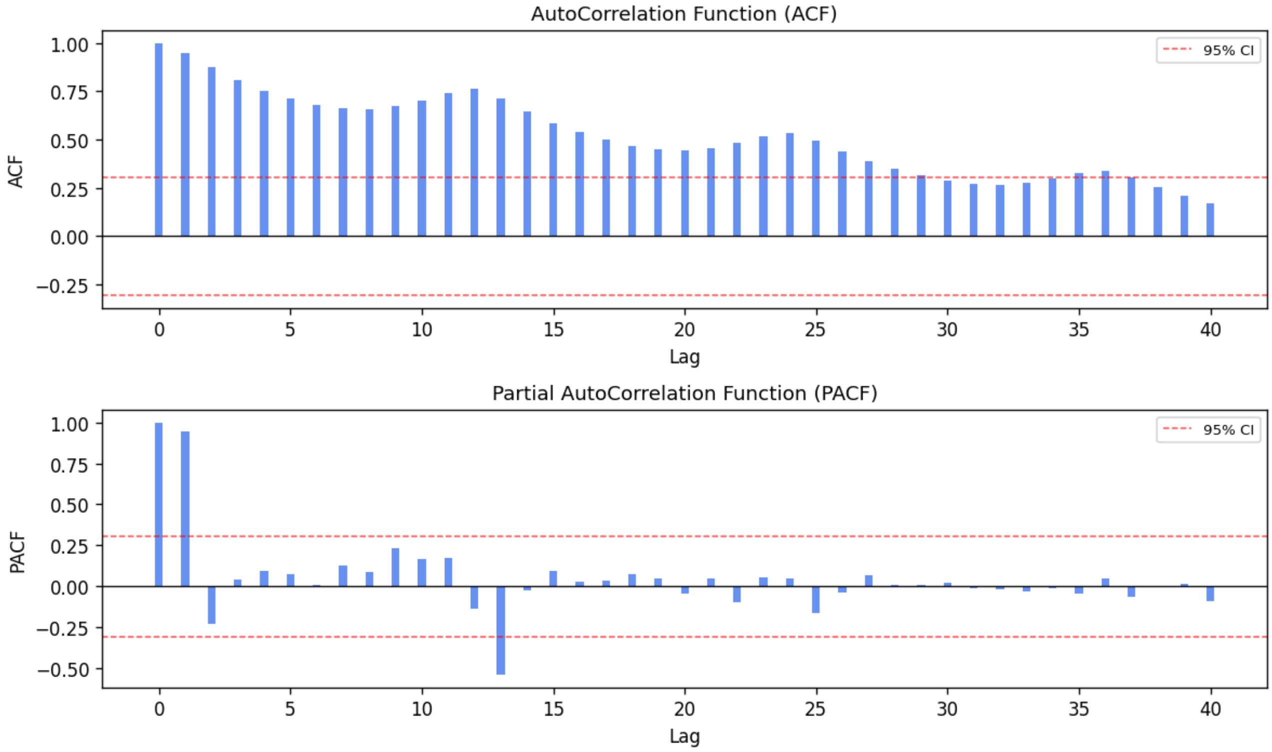
6. Forecasting Approach & Model Comparison

Selected Model: Holt-Winters

Our predictive model was chosen because it satisfies the production-forecast criteria while delivering a solid error profile—MASE 0.9597, RMSE 29.63 and MAE 22.09. Holt-Winters inherently captures both the underlying trend and the repeating seasonal pattern, which aligns with the business's need for a stable, interpretable forecast. Although SARIMA posted a slightly lower RMSE, the selection process also weighed candidate eligibility, configured loss functions, tie-breaking rules, and baseline retention, leading to Holt-Winters being retained. The decision reflects a balanced trade-off rather than a single-metric dominance, ensuring the model fits the operational constraints set for deployment.

Model	RMSE	MAE	MAPE	WAPE	MASE	Selected	Rejected Reason
Holt-Winters	29.6324	22.0884	5.76%	6.17%	0.9597	✓	
ARIMA	60.9670	46.9980	12.71%	13.13%	2.0419		Not selected (user chose a different model).
SARIMA	23.4875	19.8414	5.38%	5.54%	0.8620		Not selected (user chose a different model).
EWMA	83.5751	62.7601	15.83%	17.53%	2.7267		
Prophet	33.2324	26.4736	7.20%	7.39%	1.1502		
Dynamic Regression	40.5065	30.8449	8.02%	8.62%	1.3401		
Naive	83.5458	62.7333	15.83%	17.52%	2.7256		Baseline model for comparison.

Seasonal Naive	41.4389	37.8833	10.86%	10.58%	1.6459		Baseline model for comparison.
Mean Forecast	156.3251	143.8474	38.62%	40.18%	6.2497		Baseline model for comparison.
Drift	74.2638	53.9572	13.54%	15.07%	2.3443		Baseline model for comparison.
Holt-Winters + Auto transform	29.6324	22.0884	5.76%	6.17%	0.9597		
ARIMA + Auto transform	60.9670	46.9980	12.71%	13.13%	2.0419		
SARIMA + Auto transform	23.4875	19.8414	5.38%	5.54%	0.8620		
EWMA + Auto transform	83.5751	62.7601	15.83%	17.53%	2.7267		
Holt-Winters + Recent window	36.5464	32.1790	8.82%	8.99%	1.3981		
SARIMA + Recent window	32.4622	25.0863	6.63%	7.01%	1.0899		
EWMA + Recent window	83.5458	62.7333	15.83%	17.52%	2.7256		
Prophet + Recent window	30.7379	26.4392	7.44%	7.39%	1.1487		
Dynamic Regression + Recent window	30.0270	23.9353	6.70%	6.69%	1.0399		
Simple Ensemble	53.8262	40.9304	10.59%	11.43%	1.7783		



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7. Forecast Reliability & Performance Assessment

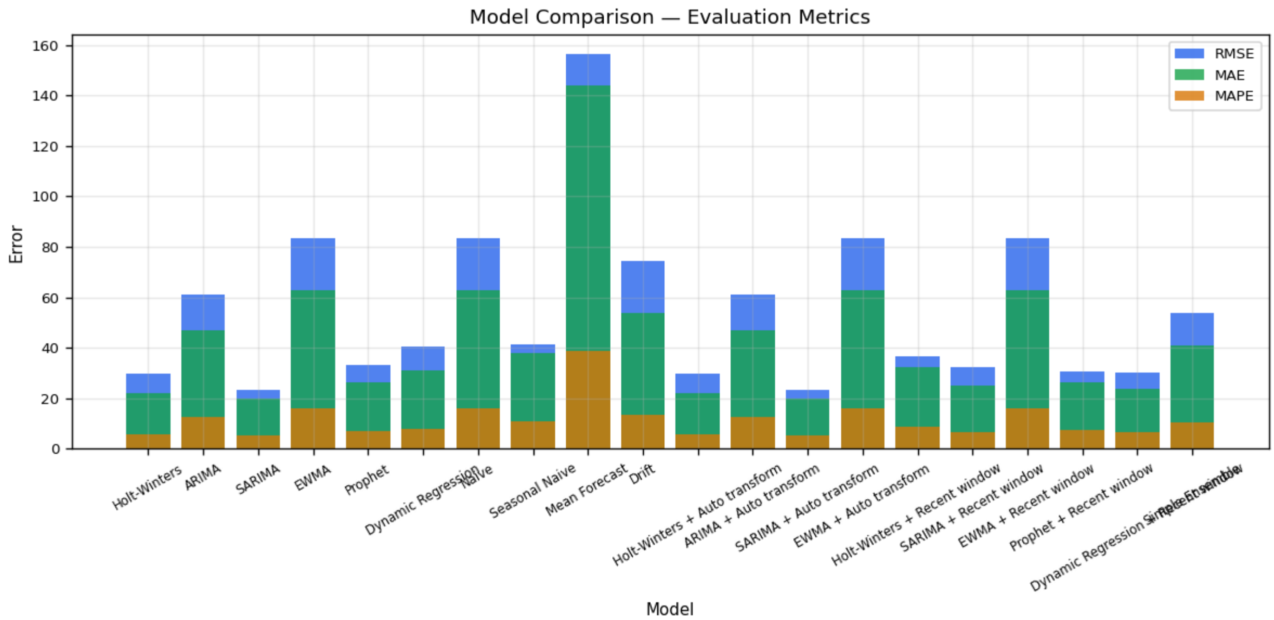
Confidence Score: 65/100 — Medium

Confidence is medium based on: Minor validation error (MAPE 5.8%); Series is non-stationary.

Forecast Health Indicators

Indicator	Status	Detail
Data Quality	Fair	Collection quality is good under the completeness and regularity policy: 0 missing values, 0 duplicate timestamps, and 0 gaps; 0 validation issues; intervals are regular. Anomaly risk requires review: 11 detected values (7.6%) exceed the 5% review threshold, limiting the overall rating to Fair.
Trend Stability	Changing	A statistically significant trend is changing the baseline over time.
Seasonality	Strong	A recurring seasonal pattern repeats every 12 periods.
Forecast Confidence	Medium	Confidence is medium based on: Minor validation error (MAPE 5.8%); Series is non-stationary.
Structural Breaks	Monitor	Candidate change points require validation of break dates, effect sizes, and persistence.
Residual Diagnostics	Concerning	Residuals are autocorrelated, indicating the model has not captured all predictable patterns.

Contributing Factors:
- Minor validation error (MAPE 5.8%)
- Series is non-stationary
- Outlier ratio 7.6% exceeds the 5% review threshold
- Detected change points may indicate a structural break
- Statistical review raised warnings



8. Explainability — Why These Conclusions

The model identifies a clear yearly cycle, with peaks and troughs repeating every 12 periods, and a steady rise of roughly 2.66 units each period, which together shape its forecasts. Recent predictions have been close to actual outcomes—within about 6 percent—demonstrating reliable performance. A lingering pattern in the remaining errors suggests the model may still be missing some information, so ongoing monitoring is advisable. Overall forecast confidence is rated 65 out of 100, indicating a moderate level of certainty.

- Strong seasonal pattern detected (every 12 periods) — The data follows a predictable recurring cycle, which the forecasting model captures to anticipate periodic peaks and troughs.
  Evidence: Seasonal period: 12
- Stable upward long-term trend identified — The data shows a consistent upward trajectory over time, which the model projects forward.
  Evidence: Trend slope: 2.657184
- Low validation error — The model's predictions were within a small margin of actual historical values, supporting its reliability.
  Evidence: MAPE: 5.76%
- Residual diagnostics require monitoring — Some predictable structure remains in the forecast errors, so model performance should be monitored.
  Evidence: Residual autocorrelation detected (Ljung-Box p=0.0014)
- Overall forecast confidence: Medium — Confidence is medium based on: Minor validation error (MAPE 5.8%); Series is non-stationary.
  Evidence: Confidence score: 65/100

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9. Statistical Audit Summary

Verdict: WARN

Statistical diagnostics confirm a pronounced monthly seasonal pattern and a well-documented set of model performance metrics, while the choice to omit outlier handling is supported by the absence of skewness or kurtosis in the series. However, checks reveal autocorrelation and a non-normal error distribution, and the identified structural break points have not been incorporated, raising concerns about the reliability of prediction intervals. A follow-up should compare alternative specifications, validate the significance and persistence of the detected breaks, and, if warranted, introduce robust error modeling or regime-specific components, while documenting the rationale and benchmarking against the SARIMA results.

Strongest Evidence:

- The detection of non-stationarity (ADF $p = 0.991$, KPSS $p = 0.01$) and the presence of a clear monthly seasonal period (12) is well-supported by the provided test statistics.
- The decision to forgo outlier transformation is justified because the profile lacks skewness/kurtosis or other distributional metrics required to select an appropriate method.
- Comprehensive reporting of candidate model metrics (RMSE, MAE, MAPE) and residual diagnostics demonstrates transparency in the forecasting pipeline.

Key Concerns:
- Model residuals are autocorrelated (Ljung-Box p-value=0.0014). The model may leave predictable patterns in the errors.
- Model residuals are not normally distributed (Shapiro-Wilk p-value=0.0348).
- Residuals fail the normality test (Shapiro-Wilk $p = 0.0348$), which may compromise prediction interval reliability.
- Change points were detected ($\text{indices} = 13, 24, 40, 61, 76, 97, 104, 112$) but were not incorporated into the forecasting model.

Recommended Follow-up:
- Review model specification and compare residual diagnostics for viable alternatives before overriding the model selected by out-of-sample accuracy.
- The calculated prediction intervals may be unreliable. This can be caused by unhandled outliers or structural breaks.
- Investigate outlier influence or consider robust error modeling.
- Validate the magnitude and persistence of these breaks and, if substantive, incorporate intervention terms, regime-specific models, or recency-weighted training windows.
- Document the rationale for the manual choice and consider presenting the superior SARIMA results as a benchmark.

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10. Strategic Risks & Operational Constraints

Historical patterns may have changed

Change-point analysis identified candidate breaks (8 possible changes) that may indicate a structural shift.

Business implication: A model fitted across differing regimes can produce misleading projections if a break is durable.

Recommended action: Validate the candidate break dates, effect sizes, and persistence first. Only if confirmed, compare intervention terms, recency weighting, segmentation, or regime-specific models.

Supporting evidence:

- Change-point count: 8

Statistical review requires follow-up

The independent statistical review raised 4 concern(s).

Business implication: Some aspects of the forecast may not be fully supported by the evidence.

Recommended action: Review the statistical audit summary and address the recommended follow-up actions.

Supporting evidence:

- Model residuals are autocorrelated (Ljung-Box p-value=0.0014). The model may leave predictable patterns in the errors.
- Model residuals are not normally distributed (Shapiro-Wilk p-value=0.0348).
- Residuals fail the normality test (Shapiro-Wilk p = 0.0348), which may compromise prediction interval reliability.

Longer-term commitments need updated forecasts

The forecast covers 12 periods at MS frequency. The horizon alone does not establish how accuracy changes across those periods.

Business implication: Longer-term commitments depend on business conditions remaining consistent with the forecast assumptions.

Recommended action: Refresh the forecast as actual results arrive and check accuracy by forecast period before making longer-term commitments.

Supporting evidence:

- Forecast horizon: 12 periods

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11. Executive Recommendations & Next Steps

1. [Medium] Continue monitoring forecast performance against future actuals to…

We will keep tracking forecast performance against future actuals to spot any drift, leveraging the completed out-of-sample validation. The model presently delivers a 5.76 % MAPE, a confidence score of 65/100, and a final-test/rolling-origin RMSE of 1.00×, reflecting solid baseline reliability. Continuous comparison of realized versus forecast values will indicate whether error stays within this validated range or a refit becomes necessary.

Evidence:
- MAPE: 5.76% (from Forecast Reliability)
- Confidence Score: 65/100 (from Forecast Reliability)
- Final-Test / Rolling-Origin RMSE: 1.00× (from Forecast Outlook)

2. [High] Validate the candidate break dates, effect sizes, and persistence

Given the high priority, we should first validate the eight candidate break dates, their effect sizes, and persistence, because change-point analysis flagged 8 potential breaks that could be transient anomalies or lasting regime shifts. If a durable break is confirmed, we can then compare alternative modeling approaches such as intervention terms, recency weighting, segmentation, or regime-specific models. This validation will determine whether a modeling adjustment is warranted and which option the evidence supports.

Evidence:
- Change Points: 8 candidate(s) (from Statistical Analysis)

3. [Medium] Align operational capacity planning with the detected 12-period sea…

Align operational capacity planning with the 12-period seasonal cycle, since historical performance shows a recurring 12-period demand pattern that creates predictable peaks and troughs. Matching capacity to this cycle lets the organization meet peak demand while avoiding excess resources during the intervening lows. As a medium-priority adjustment, this approach is expected to cover peaks without over-provisioning the trough periods.

Evidence:
- Seasonal Period: 12 (from Historical Performance)

4. [Low] Implement a rolling re-estimation of the forecast model on a MS bas…

Adopt a rolling re-estimation of the forecast model on a monthly (MS) cadence. Since the fitted trend slope is 2.657184 per period (Historical Performance), a monthly refit will capture emerging shifts before they compound across the horizon. This cadence keeps the model aligned with the latest pattern rather than relying on a fixed historical fit.

Evidence:
- Trend Slope: 2.657184 (from Historical Performance)

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12. Critical Business Assumptions

1. We assume that the economic and operational drivers that have shaped past performance will continue unchanged over the forecast horizon, and if an abrupt market shift or policy change occurs, the current baseline forecast would become obsolete.

2. The Holt-Winters approach presumes that the series’ historical level, trend and seasonal pattern remain sufficiently stable over time. Should those underlying structures shift, the model’s core assumptions would no longer hold.

3. We assume the series will maintain a stable 12-period seasonal cycle throughout the forecast horizon, and if that seasonal pattern were to shift, the model would fail to capture the new cyclical behavior.

4. The analysis rests on future data arriving at the current MS frequency, and a shift in that frequency would necessitate re-estimating the model. If the frequency changes, the model’s reliability could be compromised.

5. The model relies exclusively on the series’ own historical values and does not draw on external drivers such as competitor activity or macro-economic indicators. If this were not the case, unmodelled external forces could cause the realized outcomes to diverge materially from the forecast.

Appendix — Report Metadata

Field	Value
Engine Version	1.0.0
Generated At	2026-09-06T16:27:29.523627+00:00
Forecast Horizon	12 periods
Models Evaluated	Holt-Winters, ARIMA, SARIMA, EWMA, Prophet, Dynamic Regression, Naive, Seasonal Naive, Mean Forecast, Drift, Holt-Winters + Auto transform, ARIMA + Auto transform, SARIMA + Auto transform, EWMA + Auto transform, Holt-Winters + Recent window, SARIMA + Recent window, EWMA + Recent

window, Prophet + Recent window, Dynamic Regression + Recent window, Simple Ensemble |
Selected Model	Holt-Winters
Dataset Frequency	MS
Data Quality Rating	Fair
Row Count	144
Narrative Generation	LLM-generated